AI Evals

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What is an AI Evals?

- AI Evals (AI Evaluations) are systematic methods used to measure whether an AI system is producing correct, relevant, safe, reliable, and useful outputs.
- AI Eval = A systematic way of measuring how well an AI system performs a particular task against predefined criteria.

How agent testing is different?

- Traditional workflow based deterministic application

  assert calculate_tax(1000) == 100

- AI Agent application

  Question: What is the capital of France?

  Answer 1: Paris.

  Answer 2: The capital city of France is Paris.

  Answer 3: Paris is the capital and largest city of France.

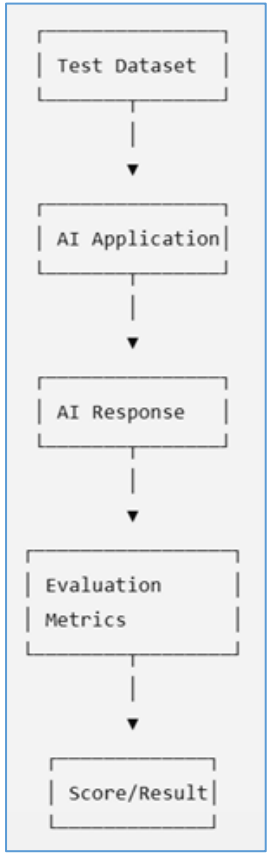
  All three may be acceptable.

  Therefore, evaluating an LLM application often requires evaluating quality, rather than simply checking whether the output exactly matches a string.

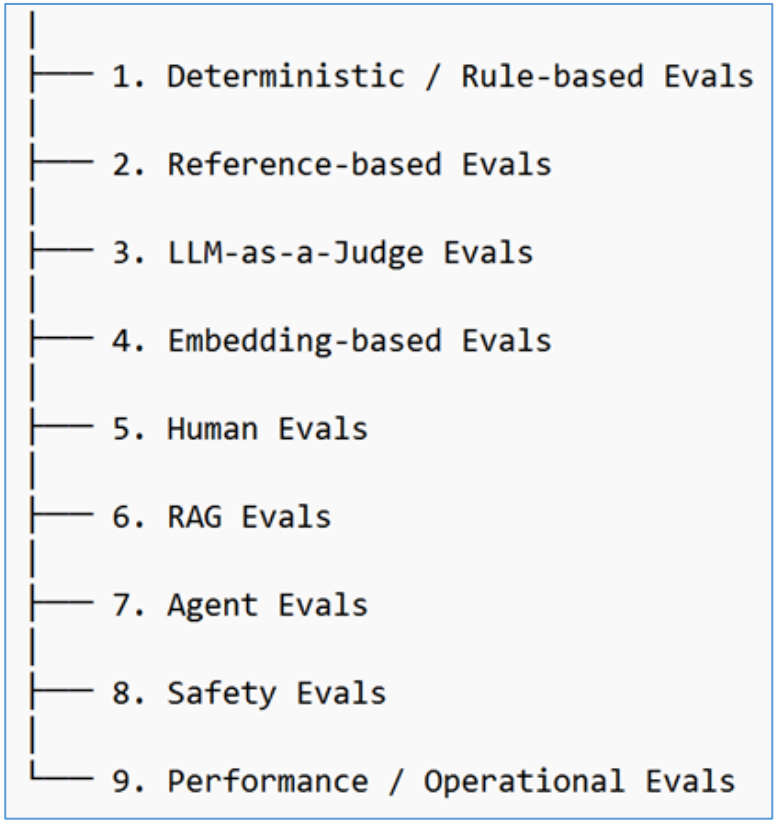
Where are AI Evals used?

Application Type	Applications
LLM based applications	Chatbot, Question answering, Summarization, Translation, Classification, Code generation
RAG based applications	Document retrieval, Context relevance, Answer faithfulness, Answer correctness
AI Agents	Tool selection, Tool arguments, Planning, Task completion
Multi-agent systems	Agent collaboration, Routing, Handoffs, Intermediate outputs

Basic AI Eval Architecture



Types of AI Evals



Types of AI Evals

- Deterministic / Rule-based Evals

```
def evaluate_response(response):
    return "Paris" in response
```

- Reference-Based Evals

```
dataset = [  {  "question": "What is the capital of France?",  "expected": "Paris"  }]
```

- LLM-as-a-Judge

Use another LLM to evaluate the output of your LLM.

- Structured LLM Evaluation

```
class Evaluation(BaseModel):
    score: int = Field(ge=1, le=5)
    reason: str
```

Types of AI Evals

- Criteria-Based Evaluation

```
class Evaluation(BaseModel):
    correctness: int
    relevance: int
    completeness: int
    clarity: int
    reason: str
```

Types of AI Evals

- Embedding-Based Evaluation

  Expected:

  "Paris is the capital of France."

  Actual:

  "France has Paris as its capital."

  Apply cosine similarity search and confirm expected output and actual output are same or not.

- Human Evaluation

  Question -> AI Answer -> Human evaluator -> 1-5 rating

Types of AI Evals

- RAG Evaluation

  User Question -> Retriever -> LLM -> Answer

  RAG Evaluation Metrics include:

  Retrieval relevance: Did we retrieve the right documents?

  Context relevance: Is the retrieved context relevant to the question?

  Context recall: Did we retrieve enough information needed to answer?

  Faithfulness/Groundedness: Is the answer actually supported by the retrieved context?

  Answer correctness: Is the final answer factually correct?

  Answer relevance: Does the answer actually answer the question?

Types of AI Evals

- **Agent Evaluation**

  Agent Evaluation has different dimensions like Task completion, Tool selection, Tool arguments, Tool execution, Planning, Number of steps, Final answer, Safety, Latency, Cost

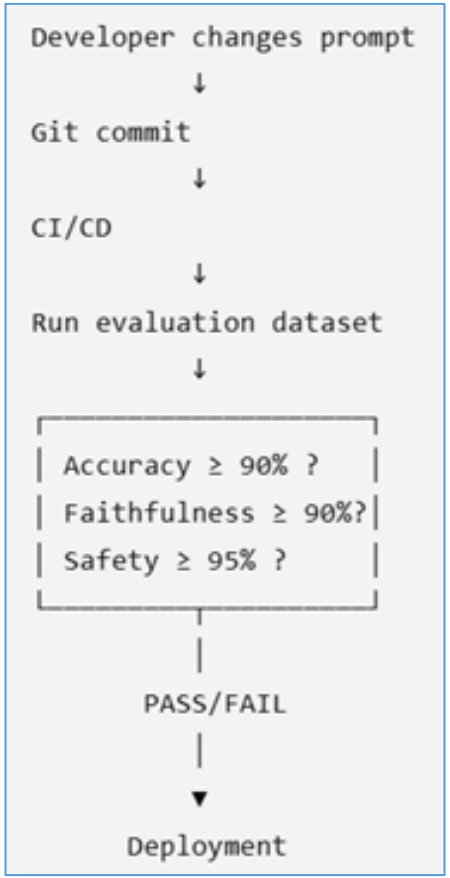
- **Safety Evals**

  AI systems can be evaluated for safety with areas like Prompt injection, PII leakage, Unsafe instructions, Jailbreak attempts, Sensitive information, Toxicity, Unauthorized actions etc.

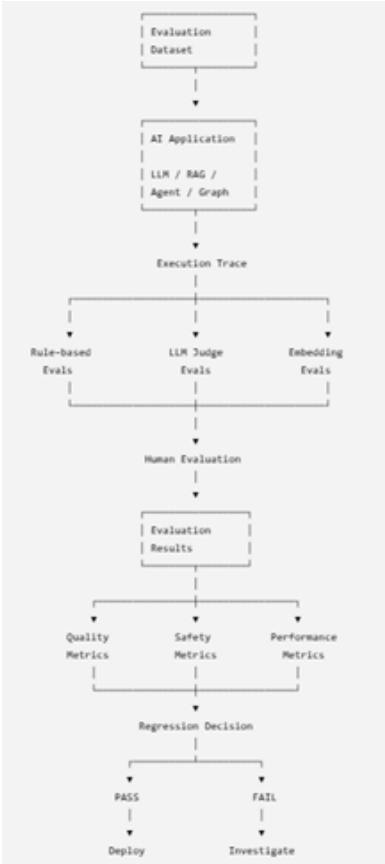
- **Performance Evals**

  AI systems can be evaluated for performance with areas like Latency, Token consumption, Cost, Throughput, Error rate, Tool calls, Number of LLM calls

Evals in CI/CD



End-to-End AI Eval Architecture



Thank you!!